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21LEM202T — Unit 5, Session 9, SLO 1

Part B — In-Class Activity

Activity 1 – Holistic Criteria Evaluation: Select one technology/system and evaluate it.

Criterion	Yes / No / Partly	Evidence / improvement needed
Real human need of body and self	**Yes**	**Solar water heater meets the real need for hot water without depleting non-renewable energy.**
Eco-friendly / cyclic / renewable	**Yes**	**Uses solar energy, which is renewable and produces no emissions during operation.**
People-friendly / mutual fulfilment	**Partly**	**Initial cost is high, limiting access for low-income families; subsidies or community models could improve this.**
User-friendly / safe / economical	**Yes**	**Safe to operate, reduces electricity bills over time, and requires minimal maintenance.**
Local resources / decentralisation / recycling	**Partly**	**Can be installed locally, but some components (e.g., glass tubes) may not be locally sourced; recycling of old panels needs improvement.**

Activity 2 – Production and Management Redesign: Answer the production questions and improve the management approach.

Question / aspect	Present answer	Holistic redesign
What to produce?	**Plastic bottled water**	**Filtered water in reusable glass bottles for local distribution**
How to produce?	**Large-scale automated bottling using fossil-fuel energy**	**Small-scale, solar-powered filtration and manual bottling with local labour**
For whom / how much?	**For urban markets, produced in excess to maximize profit**	**For the local community, produced only to meet actual demand**
Management style	**Top-down, profit-driven, with minimal worker participation**	**Cooperative, participative, with fair wages and shared decision-making**

Activity 3 – Transition Plan

Write one step at each level.

Level	Transition step
Individual transformation	**Practice self-reflection daily to align actions with natural acceptance**
Mass awareness	**Organise community workshops on the importance of harmony in relationships and nature**
Humanising education	**Integrate value-based discussions and experiential learning into the curriculum**
Holistic living model in institution / community	**Establish a model campus that demonstrates sustainable resource use and mutual respect**

Level	Transition step
Professional commitment	**Adopt ethical practices and prioritise social and environmental well-being in all professional decisions**

Part C — Long-Answer Questions

1. Why should technologies and systems be evaluated using holistic criteria? (5 marks)

Technologies and systems should be evaluated using holistic criteria because they affect not only economic efficiency but also human relationships, social well-being, and the natural environment. A narrow evaluation may lead to short-term gains but long-term harm, such as exploitation of nature or alienation among people. Holistic criteria ensure that the technology or system is aligned with universal human values like mutual fulfilment, sustainability, and harmony, which are naturally acceptable to all.

2. Explain any five criteria for holistic technologies or production systems. (5 marks)

1. **Resource efficiency**: The technology should use minimal natural resources and energy, and promote recycling and regeneration.

2. **Non-exploitative**: It should not exploit any human being or community; labour should be dignified and fairly compensated.

3. **Environmentally sustainable**: The production system should not degrade ecosystems, and should aim for zero waste and pollution.

4. **Socially inclusive**: It should benefit all stakeholders, including marginalised groups, and foster community well-being.

5. **Human-centric**: The technology should enhance human skills, creativity, and relationships rather than replace or alienate people.

3. How should a management model change to become relationship-based and mutually fulfilling? (5 marks)

A management model should shift from a hierarchical, profit-driven approach to one that prioritises trust, empathy, and mutual respect among all stakeholders. Decision-making should be participatory, with open communication and shared responsibility. The focus should move from controlling people to enabling their growth, ensuring that every individual's natural acceptance for harmony and fulfilment is respected. This includes fair wages, work-life balance, and collective ownership of goals, leading to a mutually fulfilling environment.

4. Prepare a short transition plan for value-based life and profession. (5 marks)

Individual level: Begin daily self-reflection to align thoughts and actions with natural acceptance, and practice gratitude and non-violence in all interactions.

Family level: Foster open dialogue and mutual respect at home, sharing responsibilities and resolving conflicts through understanding.

Professional level: Choose a career that serves society and nature, and apply ethical principles in every decision, prioritising collaboration over competition.

Community level: Participate in local initiatives for sustainable living, such as community gardens or waste reduction programs, and promote value-based education.

Societal level: Advocate for policies that support holistic development, such as renewable energy, fair trade, and universal access to education and healthcare.

Self-Reflection (Take-Home)

Which criterion was strongest in the system you analysed? Which was weakest?

What immediate change can you start as an individual?

What change requires institutional or community participation?

How will you use professional knowledge for mutual prosperity and preservation of nature?

Which criterion was strongest in the system you analysed? Which was weakest?

In the system I analysed, resource efficiency was the strongest criterion, as the system used renewable energy and recycled materials. The weakest criterion was social inclusivity, because the benefits were not equitably distributed among all workers.

What immediate change can you start as an individual?

I can start reducing my personal waste by carrying reusable items and avoiding single-use plastics, and also practice mindful consumption.

What change requires institutional or community participation?

Setting up a community composting facility and a local farmers' market requires collective effort from neighbours and local government support.

How will you use professional knowledge for mutual prosperity and preservation of nature?

I will apply my engineering skills to design low-cost, energy-efficient solutions for rural communities, and advocate for green technologies that create jobs while protecting ecosystems.